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Department of Computer Engineering

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Project Synopsis Template (2025-26) - Sem VII

CrptoNet

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Abstract

Cryptocurrency has become one of the most debated topics in online communities, where opinions, news, and speculation often influence market sentiment and adoption. This project applies *netnography*, a qualitative research method adapted for online environments, to study cryptocurrency-related discussions on Reddit and Twitter. By collecting and analyzing user-generated posts, comments, and tweets from selected cryptocurrency subreddits and hashtags, the study aims to identify emerging trends, community perceptions, sentiment patterns, and influential narratives. Using a combination of Natural Language Processing (NLP) techniques, sentiment analysis, and thematic categorization, the research seeks to understand how online discourse shapes public opinion and potentially impacts cryptocurrency markets. The findings are expected to provide insights into the role of social media communities in fostering awareness, speculation, and adoption of digital currencies.

Introduction

With the rise of decentralized finance, cryptocurrencies like Bitcoin, Ethereum, and other altcoins have gained global attention. While their value is partly determined by technical fundamentals and market dynamics, social media platforms have emerged as critical spaces for shaping narratives, spreading information, and influencing investor behavior. Reddit, known for its highly engaged discussion communities, and Twitter, renowned for real-time updates and viral trends, have become primary sources of cryptocurrency discourse. Netnography offers a structured way to capture and analyze these conversations, enabling deeper understanding of user sentiment, behavior, and cultural trends in the crypto ecosystem.

Problem Statement

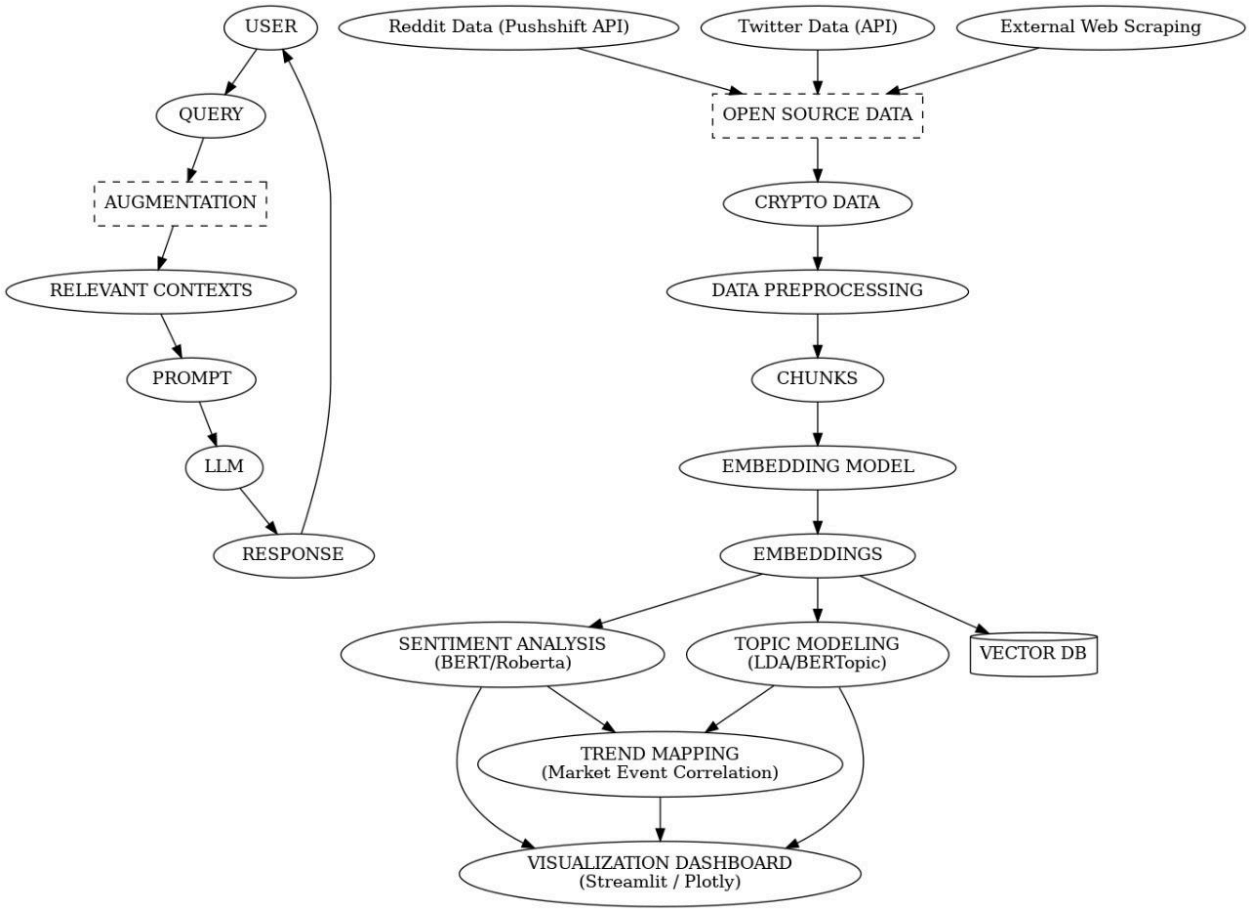
Although cryptocurrency data is widely available through market charts and price feeds, the *human perception* and *community-driven narratives* influencing those numbers are less systematically studied. The absence of structured analysis of social media conversations leads to missed insights into investor psychology, misinformation patterns, and adoption drivers. Furthermore, the fast-paced, unstructured, and jargon-heavy nature of cryptocurrency discussions makes manual tracking impractical, highlighting the need for automated, methodology-driven analysis of online discourse.

Proposed Solution

The proposed system will employ netnography combined with NLP-based sentiment and topic analysis to extract, process, and interpret cryptocurrency-related discussions from Reddit and Twitter.

Key Features:

- **Data Collection:** Automated scraping of posts, comments, and tweets from selected cryptocurrency communities and hashtags.
- **Sentiment Analysis:** Classifying discussions into positive, negative, and neutral sentiment to identify community mood shifts.
- **Thematic Categorization:** Identifying recurring topics such as investment advice, technology updates, regulatory concerns, and market speculation.
- **Trend Analysis:** Detecting patterns in discourse that correlate with significant market events.
- **Visualization Dashboard:** Presenting insights in a user-friendly dashboard for researchers and stakeholders.



Data Collection

- Extract data using APIs (Twitter API, Pushshift API for Reddit) and scraping tools.
- Filter by relevant keywords, hashtags, and subreddit names.

Data Preprocessing

- Remove duplicates, spam, and irrelevant content.
- Normalize text (lowercasing, punctuation removal, tokenization).

Sentiment & Topic Analysis

- Apply transformer-based sentiment models (e.g., BERT, RoBERTa).
- Use topic modeling (LDA / BERTopic) to extract key discussion themes.

Trend Mapping

- Correlate changes in sentiment and topics with real-world crypto market events.

Visualization & Reporting

- Develop an interactive dashboard using Plotly/Dash or Streamlit for presenting findings.

Hardware , Software and tools Requirements

Hardware:

- Processor: Intel Core i5 or above
- RAM: 8 GB minimum
- Storage: 256 GB HDD/SSD
- Operating System: Windows 11 / Linux

Software & Tools:

- Programming Language: Python
- Data Collection: Tweepy, PRAW (Python Reddit API Wrapper)
- NLP: Transformers (Hugging Face), NLTK, SpaCy
- Data Processing: Pandas, NumPy

- Visualization: Matplotlib, Seaborn, Plotly
- Topic Modeling: LDA, BERTopic
- Dashboard: Streamlit / Dash
- Database: MongoDB / PostgreSQL

Proposed Evaluation Measures

S**entiment Classification Accuracy** – Percentage of correctly identified sentiments.
Topic Modeling Coherence Score – Measures quality of extracted topics.
Trend Detection Precision – Accuracy of linking discussion trends to actual market events.
Data Coverage – Percentage of relevant data captured from selected sources.
System Response Time – Efficiency in processing and generating insights.

Conclusion

This project aims to provide a structured, data-driven analysis of cryptocurrency discourse on Reddit and Twitter using netnography and NLP techniques. The system will offer insights into sentiment shifts, key themes, and patterns that influence market perception and adoption. By bridging qualitative cultural insights with quantitative data analytics, this research will contribute to understanding the social and psychological factors that shape cryptocurrency trends.

References

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